

Growth Experiment Tracker

A CLI tool to run, track, and auto-analyze growth experiments across social, SEO, and email — with an AI agent that automatically calls **SCALE IT** or **KILL IT** on each experiment.

Philosophy: Test fast. Scale the winners. Kill the losers.

Built to explore growth engineering + AI-powered decision-making. Directly inspired by how high-velocity startups run distribution.

What it does

You log growth experiments (a LinkedIn post, an SEO article, a cold email campaign). Every day you log the result. The AI agent analyzes trends and tells you exactly what to do next.

Add experiment → Log daily results → AI analyzes → SCALE IT / KILL IT / KEEP TEST

Demo output

```
python tracker.py demo
```

```
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GROWTH EXPERIMENT ANALYSIS — 2025-05-15
=====
```

```
📌 #1 LinkedIn founder story post [SOCIAL]
Hypothesis: Personal story posts get 3x more reach than product posts
Data points: 5 | Avg: 930.0 | Latest: 2100
🚀 VERDICT: SCALE IT [high confidence]
→ Strong avg of 930 with improving trend — this is working
→ ACTION: Double the budget/frequency for next 7 days

📌 #2 SEO blog: AI for Amazon sellers [SEO]
Hypothesis: Long-tail keyword blog will rank page 1 in 30 days
Data points: 5 | Avg: 10.0 | Latest: 10
💀 VERDICT: KILL IT [high confidence]
```

- Avg of 10 is below threshold with declining trend
- ACTION: Stop this experiment and reallocate effort to winning channels

 #3 Cold email: agency owners [EMAIL]
Hypothesis: Pain-focused subject lines get >40% open rate
Data points: 5 | Avg: 43.0 | Latest: 48
 VERDICT: KEEP TESTING [medium confidence]
→ Avg of 43 – strong open rate, not enough data yet
→ ACTION: Log 3 more days of data before deciding

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SUMMARY: 1 winners | 1 killed | 3 analyzed

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Run it

```
# Install (optional – works without API key in demo mode)
pip install -r requirements.txt
export ANTHROPIC_API_KEY=your_key_here

# See demo with realistic data
python tracker.py demo

# Analyze your experiments
python tracker.py analyze

# Add your own experiment
python tracker.py add

# Log a daily result
python tracker.py log 1 450

# List all experiments
python tracker.py list
```

Commands

Command	What it does
demo	Seeds 3 realistic experiments + runs analysis

<code>analyze</code>	AI analyzes all running experiments
<code>add</code>	Add a new experiment (interactive)
<code>log <id> <value></code>	Log today's result for experiment
<code>list</code>	Show all experiments and status

Tech

- Python 3.9+ (zero dependencies for demo mode)
 - Anthropic Claude API for AI verdicts
 - Flat JSON database (no setup needed)
 - Works fully offline in demo mode
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Why I built this

I'm fascinated by growth engineering — the idea that distribution is one of the last real moats. Most founders test randomly. This tool brings the same rigor to growth that engineers bring to A/B testing in product.

The AI verdict layer means you don't need to stare at spreadsheets — the agent tells you what to do next.

Author

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